

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms a correct equation by applying conservation of energy with KE and PE – substituting the values given	AO3.4	M1	KE gained = PE lost $\frac{1}{2}mv^2 = mgh$
	Solves the equation correctly to obtain the value of v – must be rounded correctly to 2 significant figures	AO3.2a	A1	$\frac{1}{2}(75)v^2 = 75g(25)$ $v^2 = 50g$ Or $v^2 = 490$ $v = 22 \text{ m s}^{-1}$ (2sf)
(b)	Forms expressions for EPE and PE	AO3.1b	M1	PE lost = EPE gained $\frac{\lambda x^2}{2l} = mgh$
	Obtains fully correct expressions	AO1.1b	A1	
	Solves a three-term quadratic equation or substitutes correct distances for a total length of 50 m into both expressions.	AO1.1a	M1	$\frac{3200x^2}{2(25)} = 75(9.8)(25 + x)$ $64x^2 - 735x - 18375 = 0$ $x = 23.6\dots$
	Obtains the correct maximum extension of the cord or obtains the correct values for the two energies	AO1.1b	A1	$23.6 + 25 = 48.6 \text{ m}$ $48.6 \text{ m} < 50 \text{ m}$
	Compares ‘their’ results and concludes that Dominic does not get wet	AO3.2a	E1F	Dominic does not get wet
(c)	Comments that Dominic has size and considers the effect this might have on the distance fallen	AO3.2b	E1	Dominic has size so distance descended would be greater than 48.6 m
	States clearly whether Dominic does or does not get wet, justifying their conclusion	AO3.5a	E1F	He might get wet if he was taller than 1.4 m
				Total 9 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Assumes no external forces act and explicitly identifies that conservation of energy may be applied.	AO2.4	R1	Assuming no external forces act, I can use conservation of energy.
Assumes that the balls are modelled as particles or are the same size so that	AO3.3	B1	I will model both balls as particles so that the vertical distance moved is the same

they move through the same vertical distance			$\frac{1}{2} m \times v^2 = \frac{1}{2} m \times 12^2 + mg \times 1.5$ $\frac{1}{2} v^2 = \frac{1}{2} \times 12^2 + g \times 1.5$ $v^2 = 144 + 3g$ $v = \sqrt{144 + 3g}$ Hence v is independent of m so both balls will hit the ground with the same speed OR Cricket ball: $KE = \frac{1}{2} \times 0.156 \times 12^2$ $= 11.232J$ $PE = 0.156 \times g \times 1.5$ $= 2.2932J$ $\frac{1}{2} \times 0.156 v^2 = 11.232 + 2.2932$ $v^2 = 173.4$ Tennis ball: $KE = \frac{1}{2} \times 0.058 \times 12^2$ $= 4.176J$ $PE = 0.058 \times g \times 1.5 =$ $0.8526J$ $\frac{1}{2} \times 0.058 v^2 = 4.176 + 0.8526$ $= 5.0286$ $v^2 = 173.4$ Since v^2 is the same for both balls they both hit the ground with the same speed. AG
Uses KE equation and PE equation correctly for one of the balls	AO3.1b	M1	
Forms an energy equation and manipulates to find v or v^2 for one of the balls	AO1.1a	M1	
Completes a rigorous argument using energy considerations to conclude that both balls hit the ground with the same speed, either by clearly calculating values for both balls or showing that v is independent of the mass AG	AO2.1	R1	
Total 5 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an energy equation between PE and EPE	AO3.3	M1	$0.5 \times 2 \times 10 = \frac{\lambda}{2 \times 0.2} \times 0.4^2$
	Obtains correct value for λ or k	AO1.1b	A1	

	Using Hooke's Law to find extension at equilibrium	AO3.4	M1	$\lambda = \frac{10}{0.4}$ $= 25 \text{ Nm}^{-1}$ $2 \times 10 = \frac{25}{0.2} e$ $e = \frac{2 \times 10 \times 0.2}{25}$ $= 0.16$ $2 \times 10 \times 0.26 = \frac{1}{2} \times 2v^2$ $+ \frac{25 \times 0.16^2}{2 \times 0.4}$ $v^2 = 3.6$ $v = 1.897 \dots$ $= 2.0 \text{ m s}^{-1} \text{ (to 1sf)}$
	Obtains correct extension at equilibrium using 'their' λ or k	AO1.1b	A1F	
	Forms equation using conservation of energy.	AO3.4	M1	
	Obtains the correct speed. (Only accept $V = 2 \text{ m s}^{-1}$ or $V = 2.0 \text{ m s}^{-1}$)	AO1.1b	A1	
(b)	States appropriate refinement.	AO3.5c	E1	Could include air resistance.
Total 7 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses conservation of energy to form an equation.	AO3.4	M1	$\frac{1}{2} mv^2 = mg \left(\frac{r}{4} + (r - r \cos \theta) \right)$ $\frac{1}{2} v^2 = gr \left(\frac{1}{4} + 1 - \cos \theta \right)$ $v^2 = 2gr \left(\frac{5}{4} - \cos \theta \right)$ $= \frac{gr}{2} (5 - 4 \cos \theta)$ $v = \sqrt{\frac{gr}{2} (5 - 4 \cos \theta)}$ AG
	Forms correct equation.	AO1.1b	A1	
	Completes mathematical argument to find v . AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips with the change in height shown clearly and correctly.	AO2.1	R1	
(b)	Resolves radially to form	AO3.4	M1	$mg \cos \theta = \frac{m}{r} \frac{gr}{2} (5 - 4 \cos \theta)$

	equation to find θ .			
	Obtains correct value for θ from 'their' equation FT provided both M1 marks awarded	AO1.b	A1F	$2\cos\theta = 5 - 4\cos\theta$ $\cos\theta = \frac{5}{6}$ $\theta = 34^\circ$
(c)	Explains that air resistance will decrease the speed	AO3.5a	E1	Air resistance would decrease the speed and therefore increase θ
	Deduces that θ will increase	AO2.2a	R1	
(d)	Explains that the normal reaction will change as the child moves down the slope and friction is proportional to the normal reaction	AO2.4	E1	Because $F = \mu N$ and N varies as child moves down the slope
Total 8 marks				

Q5.

(a) $KE = \frac{1}{2} \times 0.16 \times 11^2$

M1

$= 9.68 \text{ J}$

A1

2

(b) Change in PE: $mgh = 0.16 \times 9.8 \times 5$

M1

$= 7.84 \text{ J}$

A1

2

(c) (i) KE when reached point B

$= 9.68 - 7.84 \text{ J}$
 $'(a)' - '(b)'$

M1

$= 1.84 \text{ J}$
 CAO

A1

2

- (ii) Speed of ball is $\sqrt{\frac{1.84}{\frac{1}{2} \times 0.16}}$
 If added in (c)(i) 0 marks for (c)(i)
 14.8 M1A1 for c(ii)

M1

$$= 4.7958 \text{ m s}^{-1}$$

$$= 4.80 \text{ m s}^{-1}$$

Condone 4.8, 4.79

A1

2

[8]

Q6.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$
 SC1 $\int_0^e \frac{\lambda e}{l} de$

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

SC1 $\int \frac{\lambda x}{l} dx$ with no limits

A1

$$= \frac{\lambda e^2}{2l}$$

A1

3

- (b) (i) Using $T = \frac{\lambda x}{l}$:

$$5g = \frac{392x}{1.6}$$

$$x = \frac{5g \times 1.6}{392}$$

$$= 0.2$$

M1

Extension is 0.2 m

A1

2

- (ii) When extension is 0.6 m, $EPE = \frac{\lambda x^2}{2l}$
B1 for 0.6

B1

$$= \frac{392 \times (0.6)^2}{2 \times 1.6}$$

M1

$$= 44.1 \text{ J}$$

A1

3

- (iii) Let y metres be distance particle is above A.
C of energy, when particle has speed 0.8 m s^{-1} , gives

$$5 \times g \times y + \frac{392 \times (0.6 - y)^2}{2 \times 1.6} + \frac{1}{2} \times 5 \times 0.8^2$$

M1 4 terms, 2 correct

M1A1 4 terms, 3 correct

M1A1

$$= \frac{392 \times (0.6)^2}{2 \times 1.6}$$

M1A2 4 terms correct

Ft answer to (b)(ii)

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A1F

$$49y + 122.5(0.6 - y)^2 + 1.6 = 122.5 \times 0.6^2$$

$$49y - 147y + 122.5y^2 + 1.6 = 0$$

$$122.5y^2 - 98y + 1.6 = 0$$

$$y = \frac{98 \pm \sqrt{98^2 - 4 \times 122.5 \times 1.6}}{2 \times 122.5}$$

$$y = \frac{98 \pm 93.9148}{245}$$

$$= 0.016674 \text{ and } 0.7833$$

if x used instead of $0.6 - y$, A1 here for $x = 0.5833...$

A1

Speed first becomes 0.8 when $y = 0.0167$

E1

5

[13]

Q7.

(a) $KE = \frac{1}{2} \times 52 \times 7^2$

M1

$= 1274 \text{ J}$

$= 1270 \text{ J}$

A1

2

(b) Change in PE: $mgh = 52 \times 9.8 \times 8$

M1

$= 4076.8$

A1

Carol's KE when she reaches the net

$= 1274 + 4076.8 \text{ J} = 5350.8 \text{ J}$

$= 5350 \text{ J}$

A1

3

(c) Speed of Carol is

$$\sqrt{\frac{5350.8}{\frac{1}{2} \times 52}}$$

M1A1

$= 14.3457 \text{ m s}^{-1}$

$= 14.3 \text{ m s}^{-1}$

A1

3

[8]

Q8.

(a) (i) Using $T = \frac{\lambda x}{l}$

Tension in string is $\frac{60 \times 2.5}{3}$
 $= 50 \text{ N}$

B1

Frictional force on A [using $F = \mu R$]
is $0.4 \times 8 \times g$
 $= 31.36 \text{ N}$

B1

which is less than tension in string
Thus particle A moves towards the hole

B1

3

- (ii) Gravitational force on B is $3g = 29.4$
which is less than tension in string

B1

Thus particle B moves towards the hole

B1

2

(b)
$$\text{EPE} = \frac{\lambda x^2}{2l}$$
$$= \frac{60 \times (2.5)^2}{2 \times 3}$$

M1

$= 62.5 \text{ J}$

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A1

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2

- (c) Let x be the distance B has moved
upwards
Work done by friction [on A] is
 31.36×0.46

M1

$= 14.4256$
 $= 14.43 \text{ J}$

A1

When B is at rest, extension is $2.04 - x$

$$\text{EPE} = \frac{\lambda x^2}{2l}$$

$$= \frac{60 \times (2.04 - x)^2}{2 \times 3}$$

$$= 10(2.04 - x)^2 \text{ J}$$

B1

C of Energy, when particle *B* is at rest, gives

$$3 \times g \times x + 10(2.04 - x)^2 + 14.4256$$

M1A1

$$= 62.5$$

A1

$$10x^2 - 11.4x - 6.4584 = 0$$

$$x = 1.555 \text{ and } -0.415$$

$$\text{Or } 10x^2 - 11.4x - 6.454 = 0$$

Particle *B* is first at rest when it has moved upwards 1.56 m

Accept 1.55

A1

7

[14]

Q9.

(a) KE at P = $\frac{1}{2} \times 25 \times 60^2$
correct

$$= 45\,000 \text{ J}$$

M1

A1

2

(b) change in PE as it falls:

$$mgh = 25 \times 9.8 \times 34$$

correct

M1

$$= 8330$$

ISW

A1

2

(c) (i) using Conservation of Energy:
KE at ground = 8330 + 45 000

ft C's (a) and (b)

M1

$$= 53\,330 \text{ J}$$

$$(\text{= } 53\,300 \text{ J to 3sf})$$

ft if M1 gained in (a) and (b)

A1

2

(ii) speed of packet is $\sqrt{\frac{1}{2} \times 25}$

ft C's (c)(i)

M1

$$= 65.3 \text{ m s}^{-1}$$

CAO

A1

2

[8]

Q10.

(a) by conservation of energy:

$$\frac{1}{2}m(u)^2 = \frac{1}{2}m(v)^2 + mg2a$$

M1 for 3 terms, 2 KE and PE; not

$$v^2 = u^2 + 2as$$

M1

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$$v^2 = u^2 - 4ag$$

A1

2

(b) (i) at point A; $T_1 = \frac{m(v)^2}{a} - mg$

both signs incorrect M1

either correct M1A1

M1A1

at point B; $T_2 = \frac{m(u)^2}{a} + mg$

both signs incorrect M1

either correct M1A1

A1

$$\frac{T_1}{T_2} = \frac{2}{5}$$

or $5T_A = 2T_B$
or $T_1 = 2T, T_2 = 5T$

B1

$$s \left(\frac{m(v)^2}{a} - mg \right) = 2 \left(\frac{m(u)^2}{a} + mg \right)$$

$$s \left(\frac{m(u^2 - 4ag)}{a} - mg \right)$$

$$= 2 \left(\frac{m(u)^2}{a} + mg \right)$$

CAO

A1

$$5u^2 - 20ag - 5ag = 2u^2 + 2ag$$

from ratio 2 : 5 or 5 : 2 and one tension equation correct

m1

$$3u^2 = 27ag$$

$$u = 3\sqrt{ag}$$

condone $\sqrt{9ag}$

A1

7

(ii) $u^2 = v^2 + 4ag \rightarrow v = \sqrt{5ag}$
condone $v^2 = 5ag$

B1

$$\text{ratio } u : v = 3 : \sqrt{5}$$

accept 1.34 : 1 or 1 : 0.745

B1

2

[11]

Q11.

(a) $KE = \frac{1}{2} \times 76 \times 28^2$

All terms correct

M1

$$= 29\,792 \text{ J}$$

$$= 29\,800 \text{ J}$$

A1

2

- (b) Change in PE: $mgh = 76 \times 9.8 \times 31 \text{ J}$

All terms correct

M1

$$= 23\,088.8 \text{ J}$$

$$= 23\,100 \text{ J}$$

A1

2

- (c) (i) KE when touches down on ground

$$= 29\,792 + 23\,088.8 \text{ J}$$

Their values, one correct

M1

$$= 52\,881 \text{ J}$$

$$= 52\,900 \text{ J}$$

CAO

A1

2

- (ii) Speed of Alan is $\sqrt{\frac{52881}{\frac{1}{2} \times 76}}$

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M1

$$= 37.304 \text{ m s}^{-1}$$

$$= 37.3 \text{ m s}^{-1}$$

CAO

A1

[8]

2

Q12.

- (a) Using conservation of energy:

$$\frac{1}{2}mv^2 = mgh$$

M1 for 2 or 3 terms, 1 KE and 1 or 2 PE

M1

$$\frac{1}{2}mv^2 = mg2.4(1 - \cos 18)$$

$$v^2 = 4.8g(1 - \cos 18)$$

$$= 2.302$$

m1A1 for finding h

m1A1

$$v = 1.52 \text{ m s}^{-1}$$

Condone 1.51

A1

4

(b) Resolving vertically:

$$T = mg + \frac{mv^2}{a}$$

Correct 3 terms

Correct signs

M1

A1

$$= 22g + \frac{22 \times 2.302}{2.4}$$

$$= 236.7... \text{ N}$$

$$= 237 \text{ N}$$

Accept 236 N

A1

3

EXAM PAPERS PRACTICE [7]

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Q13.

(a) Initial EPE = $\frac{\lambda x^2}{2l}$

$$= \frac{120 \times (0.5)^2}{2 \times 5}$$

M1 for formula with extension 0.5

M1

$$= 3 \text{ J}$$

A1

Initial KE is $\frac{1}{2} \times 0.4 \times 9^2 = 16.2 \text{ J}$

When block is at A, $\frac{1}{2}mv^2 = 3 + 16.2$

M1

$$v^2 = 19.2 \div 0.2 = 96$$

Speed is 9.80 m s^{-1}

Accept $4\sqrt{6}$; condone 9.79

A1

4

(b) (i) Normal reaction is $mg = 0.4g$

M1

Frictional force is $0.4\mu g \text{ N}$

A1

Work done by frictional force is
 $5.5 \times (0.4\mu g)$ or $2.2\mu g$

m1

C of Energy, when at A, gives

$$19.2 - 5.5 \times (0.4\mu g) = \frac{1}{2} \times 0.4 \times v^2$$

*Three terms, eg initial energy in (a)
 (=3 or 19.2); work done; KE at A.*

M1

Fully correct

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$$19.2 - 2.2\mu g = 0.2v^2$$

$$v = \sqrt{96 - 11\mu g}$$

$$Ft \ v = \sqrt{(v^2 \text{ in (a)}) - 11\mu g}$$

A1

6

(ii) Speed when rebounding is $\frac{1}{2} \sqrt{96 - 11\mu g}$

B1ft

Block is stationary at B

$$\frac{1}{2} \times 0.4 \times \frac{1}{4} (96 - 11\mu g) - 2.2\mu g$$

Three terms

M1

Two terms correct with sign

A1

$$= \frac{120 \times (0.5)^2}{2 \times 5}$$

Third term correct with sign

A1

$$\frac{1}{2} \times 0.1(96 - 11\mu g) - 2.2\mu g = 3$$

$$4.8 - 2.75\mu g = 3$$

$$\text{Or } 4.8 - 0.55\mu g - 2.2\mu g = 3$$

A1

$$\mu = 0.0668$$

A1

6

[16]

Q14.

(a) PE lost is $= 4 \times g \times 5 \cos 70 = 67.0 \text{ J}$

M1: $4 \times g \times 5 \times \cos \text{ or } \sin 20 \text{ or } 70$

M1A1

2

(b) KE is loss of PE $\Rightarrow$ KE is 67.0 J
ft

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B1

1

(c) Using KE $= \frac{1}{2} mv^2$

$$v^2 = 33.5$$

M1

Speed of particle is 5.79 m s^{-1}
(ft from (b))

A1

2

[5]

Q15.

- (a) PE is $400 \times g \times 8 = 3200 g$ [or 31 360]

B1

1

- (b) KE is $\frac{1}{2} \times 400 \times 2^2 = 800$

B1

1

- (c) Work done per minute is 32 160 J

$$\text{Power} = 32\,160 \div 60$$

$$[(a) + (b)] \div 60$$

M1

$$= 536 \text{ W}$$

CAO Accept 537 from 31 400 in (a)

A1

2

[4]

Q16.

- (a) By conservation of energy

$$\frac{1}{2}m(5v)^2 = \frac{1}{2}m(3v)^2 + mg2a$$

M1: for 3 terms, 2 KE and PE

M1

A1

$$8v^2 = 2ag$$

A1

$$v = \sqrt{\frac{ag}{4}} \text{ or } \frac{1}{2}\sqrt{ag}$$

A1

4

- (b) Greatest and least values of tension are at the highest and lowest points of its path

$$\text{At top, } T = \frac{m(3v)^2}{a} - mg$$

M1

$$= \frac{5}{4} mg$$

ft – must be positive tension

A1ft

$$\text{At B, } T = \frac{m(5v)^2}{a} + mg$$

M1

$$= \frac{29}{4} mg$$

A1ft

Ratio is 29 : 5

CAO Condone 5 : 29 or 1 : 5.8

A1

5

[9]

Q17.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$

Condone lack of limits and 'dx'

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

Must include limits from integral

A1

$$= \frac{\lambda e^2}{2l}$$

AG

A1

3

- (b) (i) Using $T = \frac{\lambda x}{l}$, $7g = \frac{196x}{2}$
M1: could use 3g or 4g – at least 1 side correct

M1

$$x = \frac{14g}{196}$$

A1

$$= 0.7$$

A1

3

- (ii) By C of Energy, when next at rest, EPE (initial)
 = PE change (for platform) + EPE (when at rest)

$$\frac{196 \times 0.7^2}{2 \times 2} = 4 \times g \times (0.7 - x) + \frac{196x^2}{2 \times 2}$$

M1: 3 terms (not including $\frac{1}{2}mv^2$)

A1: 2 of 3 terms correct

M1A1

A1: all correct

A1

$$2.45 = 2.8 - 4x + 5x^2$$

$$100x^2 - 80x + 7 = 0$$

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$$(10x - 7)(10x - 1) = 0$$

A1

$$x = 0.1$$

[last A1: must give 0.1, not 0.1 and 0.7]

A1

6

Alternative

$$\frac{196 \times 0.7^2}{2 \times 2} = 4gX + \frac{196(0.7 - X)^2}{2 \times 2}$$

(M1)

(where X is distance moved upwards)

(A1)

(A1)

$$4gX = 98 \times 0.7X + 49X^2$$

(m1)

$$X = 0, 0.6$$

(A1A1)

(iii) Max speed when $T = mg$

M1

$$4g = \frac{196x}{2}$$

A1

$$x = 0.4$$

Or mid-point of values 0.2 and 0.6 above SC2

A1

3

[15]

Q18.

(a) $KE = \frac{1}{2} \times 58 \times 2^2$

M1: Correct fully substituted expression for KE.

M1

$$= 116 \text{ J}$$

A1: CAO

A1

2

(b) Change in PE: $mgh = 58 \times 9.8 \times 7$

M1: Expression for PE with 58 and 9.8 or 9.81 with 6 or 7 for the height (or 11 and 4, 11 and 5 or 10 and 4).

M1

$$= 3978.8$$

A1: Accept 3980 or 3970 or 3978 or 3979 or 3978.8.

Accept 3982 or 3983 or 3980.

A1

$$KE = 3978.8 + 116 \text{ J}$$

M1: Adding their two previous answers.

M1

$$= 4094.8 \text{ J}$$

Speed of Kim is $\sqrt{\frac{4094.8}{\frac{1}{2} \times 58}}$

dM1: Seeing expression for v (not v^2), dependent on second M1

dM1

$$= 11.88 \text{ m s}^{-1}$$

$$= 11.9 \text{ m s}^{-1}$$

A1: Accept 11.88 or 11.8 or 11.9

Accept 11.88 or 11.8 or 11.9 or AWR 11.89 from $g = 9.81$.

A1

Obtaining $v = \sqrt{u^2 + 2gh}$ followed by incorrect substitution M0M1M1, unless h is 6 or 7, which is M1M1M1

11.0 (from $h = 6$) M1M1M1

$$v = \sqrt{2^2 + 2 \times g \times 7} \quad \text{M1M1M1}$$

$$= \sqrt{141.2} \quad \text{A1}$$

$$= 11.9 \quad \text{A1}$$

$$v = \sqrt{4 + 14g} \quad \text{M1M1M1A1}$$

$$= 11.9 \quad \text{A1}$$

$$v = \sqrt{2^2 + 12g} \quad \text{M1M1M1}$$

5

[7]

Q19.

- (a) Using conservation of energy (lowest and highest points)

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mg(2a)$$

M1: Equation for conservation of energy with two KE terms and one or two PE terms. May see m or 0.3 .

A1: Correct equation.

M1A1

$$u^2 = v^2 + 4ag$$

For complete revolutions, $v > 0$

$$\therefore u^2 > 4ag$$

$$u > 2\sqrt{ag}$$

AG

A1: Correct result with statement of $v > 0$ and some intermediate working including $4ag$ term.

A1

Or

Use of PE at top and KE at B

Correct PE and KE

Correct deduction including inequality

(M1)
(A1)
(A1)

3

(b) (i) C of Energy

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mga(1 + \sin\theta)$$

M1: Equation for conservation of energy with two KE terms and one or two PE terms including a sin θ . May see m or 0.3.

A1: Correct equation.

M1A1

$$\begin{aligned} v^2 &= \left(\sqrt{\frac{9}{2}ag} \right)^2 - 2ga(1 + \sin\theta) \\ &= \frac{5}{2}ag - 2ag \sin\theta \end{aligned}$$

Resolve radially

$$\pm R = -mg \sin\theta + \frac{mv^2}{a}$$

M1: Three term equation from resolving radially. Correct three terms, but condone signs and replacement of sin by cos.

A1: Correct equation. May see m or 0.3.

M1A1

$$= -mg \sin\theta + \frac{5}{2}mg - 2mg \sin\theta$$

$$= -3mg \sin\theta + \frac{5}{2}mg$$

$$= \left(\frac{3}{4} - \frac{9}{10} \sin\theta \right) g \quad \text{OE (must include } g \text{)}$$

A1: Simplified correct final answer.

Condone $\left(\frac{9}{10} \sin\theta - \frac{3}{4} \right) g$

A1

5

(ii) When this reaction is zero,

$$\left(\frac{3}{4} - \frac{9}{10}\sin\theta\right)g = 0$$

M1: Putting their reaction equal to zero.

M1

$$\sin\theta = \frac{5}{6}$$

θ is 56.4° above horizontal

A1: Correct angle. Accept AWRT 56.44.

A1

2

[10]

Q20.

(a)

$$\begin{aligned} \text{EPE} &= \frac{\lambda x^2}{2l} \\ &= \frac{1800 \times (4)^2}{2 \times 6} \\ &= 2400 \text{ J} \end{aligned}$$

B1: Extension = 4.

M1: Substitution of 6, 1800 and their extension into EPE formula.

A1: Correct EPE

B1

M1

A1

3

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(b)

$$\frac{1800 \times (x)^2}{2 \times 6} = \frac{1}{2} \times 200 \times 8^2$$

$$x^2 = 42.67$$

$$x = 6.53 \text{ m}$$

M1: Equation with EPE and KE terms, both correct.

$$\frac{8\sqrt{6}}{3}$$

A1: Correct extension. Accept $\frac{8\sqrt{6}}{3}$ 6.53 or AWRT 6.532.

M1

A1

Distance from O is 12.5 m

A1: Correct distance. Accept 12.5 or AWRT 12.53.

A1

3

(c) Resistance force is 800 N

Work done by resistance force is $800 \times (x + 6)$

B1: Correct work done by resistance force.

B1

C of Energy gives

$$\frac{1800 \times (x)^2}{2 \times 6} + 800 \times (x + 6) = \frac{1}{2} \times 200 \times 8^2$$

M1: Three energy terms, KE, Work Done and EPE.

A1: EPE correct.

A1: Correct equation.

(M1A1)

(A1)

$$150x^2 + 800(x + 6) = 6400$$

$$3x^2 + 16x - 32 = 0$$

$$\text{or } 150x^2 + 800x - 1600 = 0$$

A1: Correct quadratic equation with no brackets.

(A1)

$$x = \frac{-16 \pm \sqrt{16^2 + 4 \times 3 \times 32}}{2 \times 3}$$

dM1: Solving their quadratic equation with correct formula and correct substitution

dM1

$$x = 1.5497$$

A1: Correct positive solution stated.

Accept 1.54 or 1.55 or AWR 1.55.

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Distance from O is 7.55 m

A1: Correct distance from O. Accept 7.55 or 7.54 or AWR 7.55.

A1

OR

Use d for distance:

$$800 \times d$$

B1: Correct work done by resistance force.

(B1)

C of Energy gives

$$\frac{1800 \times (d - 6)^2}{2 \times 6} + 800 = \frac{1}{2} \times 200 \times 8^2$$

M1: Three energy terms, KE, Work Done and EPE.

A1: Seeing $d - 6$ in EPE

A1: EPE correct.

A1: Correct equation.

A1: Correct quadratic equation with no brackets.

(M1A1)

(A1A1)

(A1)

$$150d^2 - 1000d - 1000 = 0$$

$$3d^2 - 20d - 20 = 0$$

$$x = \frac{-20 \pm \sqrt{20^2 + 4 \times 3 \times 20}}{2 \times 3}$$

dM1: Solving their quadratic equation.

(dM1)

$$d = 7.55$$

A1: Correct distance from O. Accept 7.55 or 7.54 or AWRT 7.55.

(A1)

8

[14]

Q21.

(a) When $x \geq 22$, KE is $\frac{1}{2} \times 49 \times v^2$

EPE is $\frac{1078(x - 22)^2}{2 \times 22}$

M1 for any $\frac{1078p^2}{2 \times 22}$

M1A1

Change in PE is $49 \times g \times x$

Conservation of energy:

$$\frac{1}{2} \times 49 \times v^2 + \frac{1078(x - 22)^2}{2 \times 22} = 49 \times g \times x$$

M1 3 terms (KE, PE, EPE)

A1 2 terms correct

A1 all 3 terms correct

M1A1A1

$$\frac{49}{2}v^2 + \frac{49}{2}(x-22)^2 = 49gx$$

$$v^2 + (x-22)^2 = 19.6x$$

$$\text{SC3} \quad \frac{49}{2}v^2 + \frac{49}{2}e^2 = 49g(e+22)$$

[could use x for e]

$$5v^2 = 318x - 5x^2 - 2420$$

AG

A1

6

- (b) If x is not greater than 22, cord is not stretched

B1

1

- (c) At maximum value of x , $v = 0$

M1

$$\therefore 5x^2 - 318x + 2420 = 0$$

$$x = \frac{318 \pm \sqrt{318^2 - 4 \times 5 \times 2420}}{2 \times 5}$$

dep on M1 above

m1

$$x = 54.76... \text{ or } 8.84...$$

A1 for either solution

A1

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$$= 54.8$$

Needs to give a reason for deletion of second root. Both roots must be positive: one above 22, one below 22

E1

4

- (d) (i) When speed is a maximum, $a = 0$
tension = gravitational force

or

$$\frac{d(5v^2)}{dx} = 318 - 10x$$

M1

$$\frac{1078(x-22)}{22} = 49g$$

$$= 0 \text{ at maximum speed} \Rightarrow 318 - 10x = 0$$

A1

$$x - 22 = 9.8$$

$$x = 31.8$$

AG

A1

3

(ii) From part (a), $v^2 = 19.6 \times 31.8 - 9.8^2$

M1

$$v = 22.96$$

A1

Maximum speed is 23.0 m s^{-1}

2

[16]

Q22.

(a) Kinetic energy = $\frac{1}{2} \times 3 \times 4^2$

M1

$$= 24 \text{ (J)}$$

A1

2

(b) PE lost is $= 3 \times g \times 51$

M1

$$= 153g \text{ or } 1499.4$$

$$= 1500 \text{ J}$$

Accept 1499, 153g

A1

2

(c) (i) KE is $24 + 153g$
M1 '(a)' + '(b)'

M1

$$= 1523.4$$

$$= 1520 \text{ J}$$

(if done (c)(i) in (b) 0 marks; if done (b)

and then (c)(i) in (b) M1 only

A1

(ii) Using $KE = \frac{1}{2}mv^2$
 $v^2 = 1015.6$

M1

Speed of stone is 31.9 m s^{-1}
 Accept 31.8 from 1520

A1

*If use constant acceleration formulae in
 2D, possible 4 marks in (c) BUT **no marks**
 if initial speed is treated as being vertical*

4

- (d) eg Stone is a particle
 No air resistance

E1

1

Not no resistance; accept no wind resistance

[9]

Q23.

(a) Initial $KE = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 6 \times 12^2$

Allow one of m and v incorrect

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M1

$= 432 \text{ J}$

A1

2

- (b) (i) When it hits the ground, conservation of energy gives
 $KE = \text{Initial KE} + \text{loss in PE}$
 $= 432 + 6 \times g \times 4$
Need $6 \times g \times 4$ or 235.2

M1

$= 667.2$
 $= 667 \text{ J (3sf)}$
 AG

A1

2

(ii) $667.2 = \frac{1}{2} \times 6 \times v^2$

M1A1

Speed is 14.9 m s^{-1}

A1

3

- (iii) Stone is a particle
Not g constant

B1

No air resistance
No other forces acting

B1

2

[9]

Q24.

- (a) At maximum speed,
 tractive force = resistance force

M1

Using power = force $\times$ velocity:
 $800\,000 = F \times 40$

M1

$F = 20\,000 \text{ N}$

A1

3

- (b) Using force $\times$ distance = work done = change in energy:

$20\,000 s = \frac{1}{2} \times 60\,000 \times (40^2 - 36^2)$

M1 Fs = change of KE

M1

A1 2 of 3 terms correct

A1

A1 all 3 terms correct

A1

Distance = 456 m

A1

4

[7]

Q25.

- (a) When acceleration is zero, tension = gravitational force

$$\frac{784x}{16} = 80g$$

Both terms correct

M1

$$x = 16, x + 16 = 32 \text{ m}$$

A1 for $x = 16$

A1

Length of cord is 32 m

A1

3

- (b) (i) When bungee jumper comes to rest,

$$\text{EPE} = \frac{784 \times x^2}{2 \times 16}$$

M1

$$= \frac{49x^2}{2}$$

Change in PE = $80 \times g \times (16 + x)$

*Or $80 \times g \times 65 - (80g[16 + x])$
(or $80g(49 - x)$)*

M1

$$\frac{49x^2}{2} = 80 \times 9.8 \times (16 + x)$$

A1

$$x^2 = 32x + 512$$

$$x^2 - 32x - 512 = 0$$

AG

A1

4

(ii)
$$x = \frac{32 \pm \sqrt{32^2 + 2048}}{2}$$

M1

$$x = 43.7128$$

A1

Distance below point of jump is $43.7 + 16 = 59.7$ m
Distance between jumper and ground is $65 - 59.7$

M1

$$= 5.29 \text{ m}$$

Accept 5.287, 5.3

A1

4

[11]

Q26.

(a) $KE = \frac{1}{2} \times 55 \times 3^2$

M1

$$= 247.5 \text{ J}$$

A1

2

(b) Change in PE as slides down:

$$mgh = 55 \times 9.8 \times 20 \cos 30$$

Need $\cos 30$ or $\sin 30$

M1

$$= 9335.7...$$

A1

Using Conservation of Energy:

$$KE \text{ at end of slide} = 247.5 + 9335.7$$

'a' + '9335.7'

m1

$$= 9580 \text{ J}$$

accept 9583

A1

Speed of Anne is $\sqrt{\frac{1}{2} \times 55}$

m1

= 18.7 m s⁻¹

A1

6

- (c) Anne is a particle; no air resistance

E1

1

[9]

Q27.

(a)
$$\begin{aligned} \text{EPE} &= \frac{\lambda x^2}{2l} \\ &= \frac{180 \times 0.8^2}{2 \times 1.2} \end{aligned}$$

M1

= 48 J

A1

2

- (b) Using initial EPE = KE when string becomes slack:

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M1

$$48 = \frac{1}{2} \times 5 \times v^2$$

A1F

$$\begin{aligned} v &= \sqrt{\frac{96}{5}} \\ &= 4.38 \text{ m s}^{-1} \end{aligned}$$

$$ft \sqrt{\frac{1 \cdot a'}{2.5}}$$

A1F

3

- (c) Normal reaction is 5g or 49

M1

Frictional force is $5g \times \mu$

m1 A1

Work done by frictional force is $5\mu g \times 2$

m1

$$= 10 \mu g$$

A1

Stops at wall $\Rightarrow 10\mu g = 48$

$$m1 \ 10 \mu g = 'a'$$

m1

$$\mu = 0.490$$

accept $\frac{24}{49}$ OE

A1

7

[12]

Q28.

(a) Kinetic energy = $\frac{1}{2} \times 0.6 \times 15^2$

M1

$$= 67.5 \text{ J}$$

A1

2

(b) Using $mgh = \frac{1}{2}mv^2$:

M1

$$67.5 = 0.6 \times g \times h$$

A1

$$\begin{aligned} \Rightarrow h &= \frac{67.5}{0.6g} \\ &= 11.5 \text{ m} \end{aligned}$$

A1

3

- (c) When 3 m above ground level:
 Change in PE is $0.6 \times g \times 3$
 $= 17.64 \text{ J}$
 $\therefore$ KE of ball is $67.5 - 17.64$

M1

$$= 49.86 \text{ J}$$

A1

Speed of ball is $\sqrt{\frac{1}{2} \times 0.6 \times 49.86}$
Dep on M1

m1

$$= 12.9 \text{ m s}^{-1}$$

No KE given: speed = 12.9 SC3

A1

4

[9]

Q29.

(a) Kinetic energy = $\frac{1}{2} \times 0.6 \times 15^2$

M1

$$= 67.5 \text{ J}$$

A1

2

(b) Using $mgh = \frac{1}{2}mv^2$:

M1

$$67.5 = 0.6 \times g \times h$$

A1

$$\Rightarrow h = \frac{67.5}{0.6g}$$

$$= 11.5 \text{ m}$$

A1

3

- (c) When 3 m above ground level:

Change in PE is $0.6 \times g \times 3$
 $= 17.64 \text{ J}$
 $\therefore$ KE of ball is $67.5 - 17.64$

M1

$= 49.86 \text{ J}$

A1

Speed of ball is $\sqrt{\frac{1}{2} \times 0.6 \times 49.86}$
Dep on M1

m1

$= 12.9 \text{ m s}^{-1}$

No KE given: speed = 12.9 SC3

A1

4

- (d) eg ball is a particle, no air resistance,
 weight is the only force acting etc
Accept no spin, no wind

E1

1

[10]

Q30.

- (a) $EPE = \frac{\lambda x^2}{2l}$
 $= \frac{300 \times (1.5)^2}{2 \times 4}$

M1

$= 84.375$
 $= 84.4 \text{ J}$

A1

2

- (b) When string is slack, gain in PE is mgh
 $= 6 \times g \times 1.5 \sin 30$

M1

$= 44.1 \text{ J}$

A1

$$KE = EPE - \text{gain in PE}$$

m1

$$= 84.375 - 44.1$$

$$= 40.275$$

A1

$$\frac{1}{2} \cdot 6 \cdot v^2 = 40.275$$

$$v = 3.66$$

AG

A1

5

- (c) At A, PE gained above initial position is

$$6 \times g \times 5.5 \sin 30$$

Or PE above position string slack is 117.6

$$= 161.7 \text{ J}$$

KE at A is - 77.3

B1

This is more than initial elastic potential energy

B1

$\therefore$ particle will not reach A

E1

Or

$$\text{Using } v^2 = u^2 + 2as$$

$$a = 0.5g$$

B1

$$s = 1.37 \text{ or } 1.366$$

B1 [or 2.87 above starting point]

Hence stops before A

E1

Vertical height above sling slack is 0.683

Vertical height above starting point is 1.435

3

[10]

Q31.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

Needs limit of 0

A1

$$= \frac{\lambda x^2}{2l}$$

A1

Or

area under a straight line
= average force $\times$ distance

$$= \frac{\lambda x^2}{2l}$$

3

(b) (i) Using $T = \frac{\lambda x}{l}$
 $5g = \frac{150 \times x}{0.6}$

Extension is 0.196 m

M1

A1

2

(ii) $EPE = \frac{\lambda x^2}{2l}$
 $= \frac{150 \times (1.4)^2}{2 \times 0.6}$

M1

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$$= 245 \text{ J}$$

A1

2

(iii) When at point O ,
 $EPE = 0$
 $PE \text{ [relative to } P] = 5 \times g \times 2$

M1

$$KE \text{ [at } O] = EPE \text{ [at } P] - \text{gain in PE [at } O]$$

M1

$$\frac{1}{2}mv^2 = 245 - 10g$$

$$\frac{1}{2}.5v^2 = 147$$

A1

$$v^2 = 58.8 \quad \text{Speed is } 7.67 \text{ m s}^{-1}$$

A1

4

[11]

Q32.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

Needs limit of 0

A1

$$= \frac{\lambda e^2}{2l}$$

AG

A1

Or

area under a straight line = average force $\times$ distance $= \frac{\lambda e^2}{2l}$

3

(b) (i) Using $T = \frac{\lambda x}{l}$
 $5g = \frac{150 \times x}{0.6}$

M1

Extension is 0.196 m

A1

2

(ii) $EPE = \frac{\lambda x^2}{2l}$
 $= \frac{150 \times (0.3)^2}{2 \times 0.6}$

M1

$$= 11.25 \text{ J}$$

A1

2

(iii) When x above P ,

$$\text{EPE} = \frac{150 \times (0.3 - x)^2}{2 \times 0.6} =$$

$$\text{for } \frac{150 \times (\dots - x)^2}{2 \times 0.6}$$

M1A1

$$\text{PE [relative to } P] = (-) 5 \times g \times x$$

for $5 \times g \times \text{distance}$

M1

$$\text{KE} + \text{EPE [at new point]}$$

$$= \text{EPE [at } P] - \text{gain in PE}$$

4 terms, all signs correct, 2 terms correct

M1

$$\frac{1}{2}mv^2 + \frac{150 \times (0.3 - x)^2}{2 \times 0.6} = \frac{150 \times (0.3)^2}{2 \times 0.6} - 5gx$$

A1

$$\frac{1}{2}mv^2 + \frac{150 \times (x^2 - 0.6x)}{2 \times 0.6} = -5gx$$

Equation involving terms in v^2 , x^2 and x only

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m1

$$\frac{1}{2} \cdot 5 \cdot v^2 + 125x^2 - 75x = -49x$$

$$v^2 = 10.4x - 50x^2$$

A1

7

(iv) Particle is at rest when $v = 0$
 $10.4x - 50x^2 = 0$

M1

$$x = 0 \text{ [not required]}$$

$$\text{Or } x = \frac{10.4}{50} = 0.208 \text{ m above } P.$$

A1

2

[16]

Q33.

(a) $\frac{1}{2} \times 35 \times v^2 = 35 \times 9.8 \times 10$
Energy method

M1A1

$v = 14 \text{ (m s}^{-1}\text{)}$

A1

3

(b) Air resistance or friction

B1

1

(c) Energy lost = $35 \times 9.8 \times 10 - \frac{1}{2} \times 35 \times 12^2$ (= 910)
Difference attempted

M1

±

A1

Work done: $F \times 20$ (= 910)

m1

$20F = 910$ $F = 45.5 \text{ (N)}$

$F > 0$

A1

4

[8]

Q34.

(a) KE is loss in PE
 $= 4 \times g \times 1.5$

*M1 for $mgh = 58.8$ and then find v
without finding KE*

M1

$= 6g \text{ J or } 58.8 \text{ J}$

A1

2

- (b) When 3.5 m below O , extension is 2 m

EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(2)^2}{2 \times 1.5} = \frac{4}{3}\lambda$

M1

Change in potential energy of the particle is

$4 \times g \times 3.5$

M1

$= 14g$ or 137.2

A1

$\frac{4}{3}\lambda = 14g$

$\lambda = 102.9 \text{ N}$ or 103 N

AG

A1

4

- (c) When particle is 2.7 m below O ,

EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(2.7)^2}{2 \times 1.5} = 49.392$

Accept 49.44 [from 103]

M1A1

Change in potential energy of the particle

[from initial position] is

$4 \times g \times 2.7 = 10.8g$ or 105.84

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Conservation of energy:

$105.84 = \frac{1}{2} \times m \times v^2 + 49.392$

M1 for 3 terms and $4 \times g \times h$

M1

$2v^2 = 56.448$

Speed is $5.3126 \text{ m s}^{-1} = 5.31 \text{ m s}^{-1}$

CAO

A1

5

[11]

Q35.

(a) Kinetic energy = $\frac{1}{2} \times 5 \times 10^2$
Full method

M1

= 250 J

A1

2

- (b) Using conservation of energy:
 KE when box hits ground
 = Initial KE + Change in potential energy
Could have sign errors

M1

= 250 + 5 × 30 × g

A1ft

= 1720 J

AG; SC2 $5 \times 35.1 \times g = 1720. \dots$

A1

3

(c) $\frac{1}{2} m V^2 = 1720$

M1

$V^2 = 688$

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∴ Speed is 26.2 m s⁻¹

CAO; accept $\sqrt{688}$ or $4\sqrt{43}$; SC2 26.3

A1

3

- (d) No air resistance
Or no resistance forces

E1

Box is a particle

Deduct 1 mark for unacceptable third reason

E1

2

[10]

Q36.

(a)
$$\text{EPE is } \frac{\lambda x^2}{2l}$$
$$= \frac{200(0.5)^2}{2 \times 2}$$

M1

$$= 12.5 \text{ J}$$

A1

2

(b) When string becomes slack,

using
$$\frac{1}{2}mv^2 = \text{loss in EPE}$$

NB Using $\sqrt{5}$ to answer (a) and thus (b) $\Rightarrow$ no marks

M1

$$\frac{1}{2} \times 5 \times v^2 = 12.5$$

A1

Speed is $\sqrt{5} \text{ m s}^{-1}$
AG

A1

3

(c) Resolving vertically, $R = 5g$

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$$F = \mu R$$

M1

$$0.4 \times 5g = 2g$$

M1

Using change in energy = work done:

$$2g \times 0.5 =$$

M1 for force $\times$ distance

M1

$$\frac{1}{2} \times 5 \times (\sqrt{5}^2) - \frac{1}{2} \times 5 \times v^2$$

A1 first term (or 12.5)

A1,A1

$$9.8 = 12.5 - \frac{5}{2}v^2$$

$$v^2 = 1.08$$

Speed is 1.04 m s^{-1}

A1

7

[12]

Q37.

(a) $\frac{100}{0.4}e = 10 \times 9.8$

Use of Hookes law and equilibrium

M1

$$e = 0.392 \text{ m}$$

Correct length

A1

2

(b) $EPE = \frac{1}{2} \times \frac{100}{0.4} \times 0.6^2 = 45 \text{ J}$

Use of EPE formula

M1

AG

Correct value from correct working

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A1

2

(c) (i) $45 = \frac{1}{2} \times \frac{100}{0.4} (x - 0.4)^2 + \frac{1}{2} \times 10v^2 + 10 \times 9.8(1 - x)$

Expression for EPE with $(x \pm 0.4)^2$

M1

Correct EPE

A1

Four term energy equation

M1

$$45 = 125(x - 0.4)^2 + 5v^2 + 98(1 - x)$$

Correct GPE

B1

Correct equation

A1

$$5v^2 = 98x - 98 + 45 - 125x^2 + 100x - 20$$

Solving for v^2

dM1

$$v^2 = 39.6x - 25x^2 - 14.6 \quad \text{AG}$$

Correct result from correct working

A1

7

(ii) $39.6x - 25x^2 - 14.6 = 0$

$$25x^2 - 39.6x + 14.6 = 0$$

$$x = \frac{39.6 \pm \sqrt{39.6^2 - 4 \times 25 \times 14.6}}{2 \times 25}$$

Solving quadratic

M1

$$= 1 \text{ or } 0.584$$

Correct solutions

A1

$$x = 0.584$$

Appropriate value selected

SC Only correct answers given, award M1A1.

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A1

3

[14]

Q38.

(a) $KE = \frac{1}{2} \times 0.4 \times 8^2 = 12.8 \text{ J}$

Use of KE formula.

M1

Correct KE

A1

2

(b) (i) $KE = 12.8 + 0.4 \times 9.8 \times 6 = 36.32 \text{ J} \quad \text{AG}$

Calculation of GPE

M1

Correct KE from correct expression
(Allow use of CA equations in solutions)

A1
2

(ii) $\frac{1}{2} \times 0.4v^2 = 36.32$

Two term energy equation
Correct energy equation

M1

$$v = \sqrt{\frac{36.32 \times 2}{0.4}} = 13.5 \text{ ms}^{-1}$$

Correct speed

A1

3

- (iii) No resistance
No resistance forces
Weight is the only force
Appropriate assumption

B1

1

[8]

Q39.

(a) $\text{EPE} = \frac{1}{2} \times \frac{30}{0.5} \times 1.3^2 = 50.7 \text{ J}$

use of EPE formula

M1

correct EPE

A1

2

(b) (i) $50.7 = \frac{1}{2} \times 2v + \frac{1}{2} \times \frac{30}{0.5} \times 0.8^2$

three term energy equation
two terms correct

M1

all terms correct

A1

$$50.7 = v^2 + 19.2$$

AG

A1

solving for v

dM1

$$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$$

correct v from correct working

A1

5

$$(ii) \quad 50.7 = \frac{1}{2} \times 2v^2$$

M1

two term energy equation

correct equation

A1

$$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$$

correct velocity

A1

3

$$(c) \quad \frac{1}{2} \times 2v^2 = 50.7 - 1.8 \times 0.1 \times 2 \times 9.8$$

finding friction force

M1

correct friction force

A1

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three term energy equation

M1

correct equation

A1

$$v = \sqrt{47.172} = 6.87 \text{ ms}^{-1}$$

correct velocity

A1

5

[15]

Q40.

$$(a) \quad KE = \frac{1}{2} \times 0.6 \times 14^2 = 58.8 \text{ J}$$

use of KE formula

M1

correct energy

A1

2

(b) $0.6 \times 9.8h = 58.8$

two term energy equation involving PE and previous energy

M1

$$h = \frac{58.8}{0.6 \times 9.8} = 10 \text{ m}$$

correct equation

A1

correct height

A1

3

Note: Constant acceleration methods not accepted.

- (c) (i) WD against resistance
three term energy equation

M1

$$= 58.8 - 0.6 \times 9.8 \times 8$$

correct equation

A1

$$= 11.76 = 11.8 \text{ J (to 3 sf)}$$

correct value

A1

3

(ii) $8F = 11.76$

using work done = Fd with $d = 8$

M1

$$F = 1.47 \text{ N}$$

correct force

accept 1.48

A1ft

2

- (d) The magnitude of the force would vary with the speed of the ball.
appropriate explanation

B1

1

[11]

Q41.

(a) $12.5 = \lambda \times \frac{0.1}{0.4}$

M1: subs. A1: All correct

M1A1

$\lambda = 50$

A1

3

(b) $EPE = \frac{50 \times (0.1)^2}{2 \times 0.4}$
Subs.

$= 0.625 \text{ J}$

PI A1: all correct

M1

A1

$0.625 = \frac{1}{2} \times 0.2 \times v^2$

M1 use of principle

M1

ft EPE

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A1F

$v = 2.5 \text{ ms}^{-1}$

ft EPE

A1F

5

[8]

Q42.

(a) (i) $\frac{1}{2} \times 28 \times 1^2 + 28 \times 9.8 \times 2.5 = \frac{1}{2} \times 28 \times v^2$

M1 all 3 terms – 1 each term correct

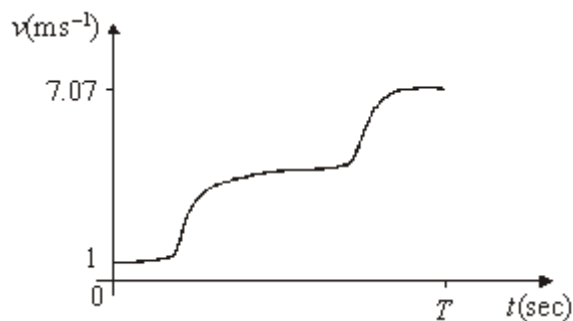
M1A2

$v = 7.07 \text{ ms}^{-1} \text{ (3 sf)}$

A1

4

(ii)



v increasing

B1

accept straight line, not horizontal
labels all correct (1, 7.07, T)

B1

correct shape

B1

3

(b) Initial energy = PE + KE

$$\frac{1}{2} \times 28 \times 1 + 28 \times 9.8 \times 2.5$$

M1

$$700 - \frac{1}{2} \times 28 \times v^2 = 350$$

M1 work/energy principle

A1 correct

M1A1

$$v = 5 \text{ ms}^{-1}$$

ft slip eg sign

A1F

4

[11]

Q43.

(a) KE = Initial PE

Alternative for (a):

$$\frac{1}{2} (50) v^2 = (50) g (20)$$

Use of $v^2 = u^2 + 2as$

M1

$$\therefore v^2 = 40g$$

$$v^2 = 0^2 + 2g (20)$$

$$v \approx 19.8 \text{ ms}^{-1}$$

$$v = 19.8 \text{ ms}^{-1}$$

A1

2

(b) (i) EPE after stretching = PE at start

M1

$$= 50g (32)$$

$$= 15\,680 \text{ J}$$

Alternative for (b)(i):

EPE after stretching = PE + KE at natural length

$$= 50 (g) (12) + \frac{1}{2} (50) (19.8..)^2$$

$$\text{AG} = 15680 \text{ J}$$

A1

2

(ii) $\frac{1}{2} k (32 - 20)^2 = 15\,680$

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B1 for $\frac{1}{2} k x^2$; M1 for equation

M1B1

$$72k = 15\,680$$

$$k = 218$$

A1 CAO

A1

3

[7]

Q44.

(a) use of mgh

M1

$$60(9.8)(20 - 20 \cos 60^\circ)$$

$20 \cos 60^\circ$ seen

B1

$= 5880 \text{ J}$

A1

3

(b) (i) energy at M = energy at P

$$\frac{1}{2} mu^2 + mgh = \frac{1}{2} mv^2$$

$$\frac{1}{2} (60)(6)^2 + 60(9.8)(20 - 20 \cos \theta)$$

M1 one term correct

A1 two terms correct

M1

A1

$$= \frac{1}{2} (60)v^2$$

fully correct

A1

$$18 + 196 - 196 \cos \theta = \frac{1}{2} v^2$$

$$v^2 = 428 - 392 \cos \theta$$

AG

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A1

4

(ii) $mg \cos \theta - N = \frac{mv^2}{r}$

$$\frac{mv^2}{r} \text{ used or seen}$$

B1

attempt at Newton's law

M1

$$60(9.8) \cos \theta - N = \frac{60}{20} (428 - 392 \cos \theta)$$

substitute v^2

A1

$$N = 588 \cos \theta - 1284 + 1176 \cos \theta$$

$$\text{or } 1764 \cos \theta - 1284$$

can be unsimplified – must be $N = \dots$

A1

4

(iii) $N > 0$ from M to N

$$1764 \cos \theta - 1284 > 0$$

solve equation or inequality

M1

$$\therefore \cos \theta > \frac{1284}{1764} \approx 0.72789$$

$$\theta < 43.2\dots$$

finding an angle $< 90^\circ$

A1ft

loses contact since $43^\circ < 60^\circ$

correct interpretation

A1

3

Alternative method:

when $\theta = 60^\circ$

evaluate N at end of bridge

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(M1)

$$N = -402 < 0$$

Their N evaluated

(A1ft)

loses contact before end of bridge

correct interpretation

(A1)

[14]

Q45.

(a)
$$\text{EPE} = \frac{1}{2} (50)(0.03)^2$$

Use of correct formula

M1

$$= 0.0225 \text{ J}$$

A1

2

(b) $\text{KE} + \text{PE} = \text{EPE}$

$$\frac{1}{2}mv^2 + mgh = \text{EPE}$$

KE or PE term correct

B1

$$\frac{1}{2}(0.02)v^2 + (0.02)(9.8)(0.03) = 0.0225$$

Equation formed

Equation correct; ft EPE

M1 A1f.t.

$$v \approx 1.29 \text{ ms}^{-1}$$

AG

A1

4

- (c) Energy remains constant –
No KE at end, no EPE at end

$$\therefore (0.02)(9.8)h = 0.0225$$

M1A1

$$\therefore h = 0.11 \text{ m}$$

ft EPE

2 sig fig accuracy

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A1f.t.

3

i.e. 11 cm

Alternatives to part (c)

Energy remains constant

KE at point of release = PE on reaching max height

$$\frac{1}{2}(0.02)(1.29)^2 = (0.02)gh$$

(M1)

$$\therefore h = 0.849\dots$$

(A1)

$$\therefore \text{height above initial position} = 0.849\dots + 3 = 0.11\text{m}$$

ft their h value + 3

(A1f.t.)

or Use of $v^2 - u^2 = 2as$

(M1)(A1)

$s = 0.849$ distance = $0.849 + 3$

ft their $(s + 3)$ total

(A1f.t.)

[9]

Q46.

(a) PE at A = KE at B

$$\frac{1}{2} mu^2 = mgr$$

Attempt to use conservation of energy

M1

$$\Rightarrow u^2 = 2gr$$

A1

2

Alternative

Use $v^2 - u^2 = 2as$

(M1)

$$v^2 - 0^2 = 2gr$$

$$u^2 = 2gr$$

(A1)

(b) (i) Conservation of energy

Energy at B = Energy at P

$$mu^2 + mgr \sin \theta = \frac{1}{2} mv^2$$

Attempt to use conservation of energy

M1

A1 – one term correct

A1

A1 – all correct

A1

3

$$v^2 = 2gr(1 + \sin \theta)$$

Use of $u^2 = 2gr$

Printed answer

(A1)

Alternative

Energy at A = Energy at P

$$mgr + mgr \sin \theta = \frac{1}{2} mv^2$$

Attempt to use conservation of energy

(M1)

A1 – one term correct

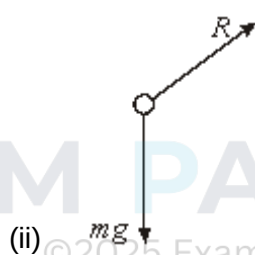
(A1)

A1 – all correct

(A1)

Printed answer

(A1)



Use mg or W

B1

1

(iii) Res force to centre

$$= R - mg \sin \theta$$

Seen in equation or stated

B1

$F = ma$ radially gives,

$$R - mg \sin \theta = \frac{mv^2}{r}$$

Follow through error above

M1A1f.t.

$$R = 2mg(1 + \sin \theta) + mg \sin \theta \text{ or } mg(2 + 3\sin \theta)$$

Use of v^2 to find R , need not be simplified

A1

4

(c) $\theta = 90^\circ$, $R = 5mg$

Either their R with $\theta = 90^\circ$

or their $R - mg$ without $\theta = 90^\circ$

M1

Reduces to mg after C since no longer circular motion.

$\therefore$ 4mg reduction

Fully explained for A1

A1

2

(d) $u^2 = 4gr$

Use of (b)(i) to find u^2

B1

Use $v^2 - u^2 = 2as$

$$0^2 - 4gr = 2a(4r)$$

Suitable equation

M1

$$\text{acc} = \frac{-g}{2}$$

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$$\text{retardation} = \frac{g}{2}$$

Condone +/-

A1

3

[16]

Q47.

(a) Rope taut (momentarily) after 14m,

so PE = KE

$$70g(14) = \frac{1}{2} (70)v^2$$

attempt at conservation of energy

$$\text{or } v^2 = u^2 + 2as$$

M1

$$v^2 = 28g$$

$$v \approx 16.6$$

AWRT

A1

2

- (b) (i) Initial PE = EPE at end

$$70g(14 + x) = \frac{1}{2}(196)x^2$$

attempt at conservation of energy equation

fully correct

For EPE (any stage)

M1 A1 B1

$$70(14 + x) = 10x^2$$

$$7(14 + x) = x^2$$

$$0 = x^2 - 7x - 98$$

Correct quadratic – AG

A1

4

- (ii) $0 = (x - 14)(x + 7)$

any valid method for solving quadratic

M1

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$$x = 14 \text{ or } -7$$

$$\therefore x = 14 \text{ m}$$

A1

2

- (iii) Hooke's law: $T = kx$

$$\therefore T = 196(14)$$

Hooke's Law used; ft their x

B1f.t.

$$\text{Newton's law: } 70g - T = 70a$$

M1 attempt $F = ma$

M1A1

$$\therefore \frac{70(9.8) - 196(14)}{70} = a$$

$$-29.4 \text{ ms}^{-2}$$

accept + or –

A1

4

[12]

- (a) Rope taut (momentarily) after 14m,

so PE = KE

$$70g(14) = \frac{1}{2} (70)v^2$$

attempt at conservation of energy

or $v^2 = u^2 + 2as$

M1

$$v^2 = 28g$$

$$v \approx 16.6$$

AWRT

A1

2

- (b) (i) Initial PE = EPE at end

$$70g(14 + x) = \frac{1}{2} (196)x^2$$

attempt at conservation of energy equation

fully correct

For EPE (any stage)

M1 A1 B1

$$70(14 + x) = 10x^2$$

$$7(14 + x) = x^2$$

$$0 = x^2 - 7x - 98$$

Correct quadratic – AG

A1

4

- (ii) $0 = (x - 14)(x + 7)$

any valid method for solving quadratic

M1

$$x = 14 \text{ or } -7$$

$$\therefore x = 14 \text{ m}$$

A1

2

(iii) Hooke's law: $T = kx$

$$\therefore T = 196(14)$$

Hooke's Law used; ft their x

B1f.t.

Newton's law: $70g - T = 70a$

M1 attempt $F = ma$

M1A1

$$\therefore \frac{70(9.8) - 196(14)}{70} = a$$

$$-29.4 \text{ ms}^{-2}$$

accept + or -

A1

4

[12]

Q48.

(a) At C, $N = 0$, so only force on the car will be mg

Stated or implied

B1

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$$\text{Newton's law radially: } mg = \frac{mv^2}{r}$$

if " $\cos \theta$ " used then must state value of $\theta + c$

M1

$$v^2 = gr$$

AG

A1

3

(b) Using conservation of energy between B and C:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + 2mgr$$

Any KE/Pe term seen



M1 equations formed, clear indication of position being considered

B1 M1A1

$$u^2 = 5gr$$

Substitute v^2 and rearrange

A1

4

- (c) Using conservation of energy between A and B:

$$mgh = \frac{1}{2}mu^2$$

PE = KE

M1

$$\Rightarrow mgh = \frac{1}{2}m5gr$$

ft their u^2

A1f.t.

$$h = 2.5r$$

$$\Rightarrow k = 2.5$$

A1

3

- (d) Conservative forces/friction

E1

1

[11]

Q49.

$$(a) \quad EPE = \frac{1}{2} \times \frac{40}{2} \times 3^2 = 90 \text{ J}$$

Finding EPE

M1

ag Correct EPE from correct working

A1

2

$$(b) \quad 90 = \frac{1}{2} \times 5v^2$$

Use of EPE = KE

M1

Correct equation

A1

$$v^2 = 36$$

$$v = 6$$

ag *Correct speed from correct working*

A1

3

(c) $EPE = \frac{1}{2} \times \frac{40}{2} \times 1^2 = 10 \text{ J}$

Finding EPE 3 metres from O

M1

Correct EPE

A1

$$90 - 10 = \frac{1}{2} \times 5v^2$$

Using EPE lost = KE

M1

Correct equation

A1

$$v^2 = 32$$

$$v = 5.66 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Correct speed

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A1

5

[10]

Q50.

(a) (i) $KE = \frac{1}{2} \times 35 \times 2^2 = 70 \text{ J}$

KE calculation using $v = 2$

M1

Correct KE

A1

2

(ii) $\text{Gain in KE} = \frac{1}{2} \times 35 \times 6^2 - 70$

KE calculation using $v = 6$

M1

Correct expression for gain in KE

A1

$$= 560 \text{ J}$$

Correct gain

A1

3

(b) $\text{PE lost} = 35 \times 9.8 \times 10 \sin 40^\circ$

PE calculation with attempt to find height

M1

$$= 2200 \text{ J}$$

ag *Correct PE; allow 2204*

A1

2

(c) $2204 - 560 = 10F$

Energy lost = Fx including 560

M1

Correct equation

A1

$$F = 164 \text{ N}$$

Correct F

Alternative

M1: Three term equation of motion

A1: Correct equation

A1: Correct force

A1

3

(d) $\frac{1}{2} \times 35 \times 6^2 = Fs$

Use of $\text{KE lost} = Fx$

M1

Correct equation

A1ft

$$s = \frac{630}{F} = 3.83 \text{ m}$$

Correct length

Follow through

F Allow 3.84

Alternative

M1: Finding acceleration

A1: Correct use of a constant acceleration equation

A1: Correct length from correct working

A1ft

3

[13]

Q51.

(a)
$$\text{EPE} = \frac{30 \times 0.8^2}{2 \times 2} = 4.8 \text{ J}$$

Use of EPE formula with 0.8

Correct EPE

M1

A1

2

(b) (i)
$$4.8 = 0.15 \times 9.8 \times 2.8 + \frac{1}{2} \times 0.15 \times v^2$$

Three term energy equation

Accept $0.684 = \frac{1}{2} 0.15 v^2$

M1

$$v = \sqrt{\frac{4.8 - 4.116}{0.075}} = 3.02 \text{ ms}^{-1}$$

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Correct equation

A1

Solving for v

m1

ag Correct v from correct working

A1

4

(ii)
$$4.8 = 0.15 \times 9.8 \times 2.8 + 0.15 \times 9.8h$$

Three term energy equation using height above O

Accept $0.684 = mgh$

M1

Correct equation

A1

$$h = \frac{4.8 - 4.116}{1.47} = 0.465 \text{ m}$$

Correct height above O

Accept 0.47 or 0.46

A1

As $0.465 < 2$ the string does not become taut

Correct conclusion

Alternative

M1: Use of constant acceleration equation

A1: Correct equation

A1: Correct height

A1: Correct conclusion

A1

4

[10]

Q52.

(a) Using conservation of energy

$$\frac{1}{2}mv^2 = mga(1 - \cos \theta)$$

M1 A1

$$v^2 = 2ga(1 - \cos \theta)$$

$$v = \sqrt{2ga(1 - \cos \theta)}$$

A1

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3

(b) Particle leaves the surface when

$$mg \cos \theta = \frac{mv^2}{a}$$

M1 A1

$$mg \cos \theta = \frac{m}{a} \times 2ga(1 - \cos \theta)$$

M1

$$\cos \theta = 2 - 2 \cos \theta$$

$$\cos \theta = \frac{2}{3}$$

A1

4

[7]

Q53.

(a) (i) Initial KE = $\frac{1}{2} \times 65 \times 2^2$
Use of KE formula

M1

= 130 J
Correct energy

A1

2

(ii) $65 \times 9.8h = 130$
Using $mgh = 130$

M1

$h = \frac{130}{637} = 0.204 \text{ m (to 3sf)}$
Correct h

A1

2

(b) (i) KE = $130 + 65 \times 9.8 \times 6 = 3950 \text{ J (3 sf)}$
Sum of KE + PE or PE at $h = 6.204$
Correct equation
Correct energy

M1 A1 A1

3

(ii) $\frac{1}{2} \times 65v^2 = 3952$
Use of KE formula to find v

M1

$v = \sqrt{121.6} = 11.0 \text{ ms}^{-1}$
Correct v from their energy in part (b)(ii)

A1ft

2

[9]

Q54.

(a) $20 \times 9.8 = \frac{0.7\lambda}{2}$
Use of $T = mg$

M1 A1

$\lambda = \frac{2 \times 20 \times 9.8}{0.7} = 560$
Correct result from working

A1

3

(b) (i) $20 \times 9.8L = \frac{560(L-2)^2}{2 \times 2}$
Two term energy equation
Correct terms
Correct signs

M1 A1 A1

$196L = 140L^2 - 560L + 560$
Expanding and simplifying

m1

$5L^2 - 27L + 20 = 0$
Correct result from correct working

A1

5

(ii) $L = \frac{27 \pm \sqrt{27^2 - 4 \times 5 \times 20}}{2 \times 5}$
Solving a quadratic

M1

$= 4.51 \text{ or } 0.886$
Correct solutions

A1

$L = 4.51$
Selecting the appropriate solution

A1

3

[11]

Q55.

(a) (i) $KE = 2 \times 9.8 \times 4 = 78.4 \text{ J}$

Use of $KE = \text{change in } PE \text{ with } h = 4$

M1

Correct energy

A1

2

(ii) $78.4 = \frac{1}{2} \times 2 \times v^2$

Use of kinetic energy or constant acceleration formula
to form an equation in v based on a fall of 4 metres

M1

Correct equation

A1

$v = \sqrt{78.4} = 8.85 \text{ ms}^{-1}$

Correct v

A1

3

(b) (i) $78.4 + 19.6x = \frac{80}{2 \times 4} x^2$

Calculation of EPE shown

M1

Correct EPE

A1

Three term energy equation

M1

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$0 = 10x^2 - 19.6x - 78.4$

ag Correct equation from correct working

A1

4

(ii) $x = \frac{19.6 \pm \sqrt{19.6^2 - 4 \times 10 \times (-78.4)}}{2 \times 10}$

Solving the quadratic equation

M1

$= 3.95 \text{ or } -1.99$

Correct solutions

A1

Max $L = 7.95 \text{ m}$

A1ft
3

- (c) No air resistance
Appropriate assumption

B1
1

[13]

Q56.

(a) $KE = \frac{1}{2} \times 0.3 \times 18^2 = 48.6 \text{ J}$
Calculating KE

M1

Correct KE

A1
2

(b) $48.6 = 0.3 \times 9.8h$
Forming equation and solving for h

M1

$$h = \frac{48.6}{2.94} = 16.5 \text{ m}$$

Correct h

A1
2

(c) $48.6 = 0.3 \times 9.8 \times 2 + KE$

$$KE = 48.6 - 5.88 = 42.7 \text{ J}$$

Calculating PE gained

M1

Correct PE

A1

Correct KE

A1

$$42.72 = \frac{1}{2} \times 0.3v^2$$

$$v = \sqrt{\frac{42.72}{0.15}} = 16.9 \text{ ms}^{-1}$$



m1

Correct v

A1

5

[9]

Q57.

$$(a) \quad EPE = \frac{12 \times 1^2}{2 \times 0.5} = 12 \text{ J}$$

Calculating EPE

M1

Correct EPE

A1

2

$$(b) \quad (i) \quad KE = 12 - 0.5 \times 9.8 \times 1$$

Using PE to obtain KE

M1

$$= 7.1 \text{ J}$$

Correct answer from correct working

A1

2

$$(ii) \quad KE = 7.1 - 0.5 \times 9.8 \times 1$$

Using PE to obtain KE

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$$= 2.2 \text{ J}$$

Correct answer

A1

2

$$(iii) \quad 0.5 \times 9.8x + \frac{12x^2}{2 \times 0.5} = 2.2$$

Correct EPE

B1

Three term energy equation

M1

Correct equation

A1

$$12x^2 + 4.9x - 2.2 = 0$$

$$x = \frac{-4.9 \pm \sqrt{4.9^2 - 4 \times 12 \times (-2.2)}}{2 \times 12}$$

Solving quadratic

m1

$$x = 0.270 \text{ or } -0.679$$

Correct solutions

A1

$$\text{Max height} = 0.5 + 0.270 = 0.770\text{m}$$

Calculating actual height

m1

Correct height

A1

7

[13]

Q58.

- (a) (i) $\text{KE} = 0.5 \times 3 \times 9.8 = 14.7\text{J}$
KE based on a fall of 3 m
cao

M1 A1

2

- (ii) $\text{KE} = 0.5 \times 6 \times 9.8 = 29.4\text{J}$
Using 29.4

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B1

$$\frac{1}{2} \times 0.5v^2 = 29.4$$

Forming and solving an energy equation

M1

$$v = \sqrt{\frac{29.4}{0.25}} = 10.8 \text{ ms}^{-1}$$

awrt 10.8

A1

3

- (b) $\frac{1}{2} \times 0.5 \times 2^2 = 29.4 - 6R$

Three term energy equation with 6R

Correct equation

M1 A1

$$R = \frac{29.4 - 1}{6} = 4.73 \text{ N}$$

Solving for R
awrt 4.73

m1 A1

4

[9]



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Examiner reports

Q1.

In this question, students were able to demonstrate their understanding of principles of energy shown by a variety of correct approaches in part (b). The best approaches had clearly labelled diagrams which helped students obtain correct and consistent expressions for different types of energies. The mean mark for part (a) was 51%, the mean mark for (b) was 43% and the mean mark for (c) was 19%.

Part (a) was meant to be a simple application of conservation of energy when the cord reached its natural length. Many students applied the formulae correctly but did not give the final answer to two significant figures, thus losing the final mark. Required accuracy should match the value of g given in the question. A less common error was to use the height of the bridge above the river rather than the 25 m required.

There were many different approaches to part (b) and if students had used diagrams they would have been less confused and obtained more marks. The approaches seen were:

- calculation of the initial total energy (36750 J) and to compare this with the energy required for Dominic to reach the river (40000 J), deducing that there was insufficient energy to do so
- forming an energy equation and attempting to find the speed of Dominic at the surface of the river, deducing that he did not reach the river as v^2 was negative
- using x as the maximum extension, forming an energy equation to deduce that $x = 23.6$ m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the maximum length, forming an energy equation to deduce that $x = 48.6$ m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the minimum distance above the river, forming an energy equation to deduce that $x = 1.4$ m deducing that Dominic would not get wet.

All these methods proved to be successful for some students. However, without clear diagrams other students often got confused about what they were finding, leading to incorrect combinations of the different types of energies. Sometimes an x that had started out as a maximum length became an extension instead.

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Part (c) linked to part (b) and required students to realise that if Dominic was not a particle then he would have height and hence could get wet if he was taller than 1.4 m. Any reference to air resistance here scored zero marks.

Q5.

Parts (a) and (b) were usually answered correctly but, in part (c)(i), a number of students added the two energies rather than subtracting them. Part (c)(ii) was usually completed correctly.

Q6.

Many students were unable to show convincingly that the work done in stretching the string was $\frac{\lambda e^2}{2l}$. An integration using a variable, usually x , was required. A number of

students were penalised for using either $\int_0^e \frac{\lambda e}{l} de$ or $\int_0^{\lambda x} \frac{\lambda x}{l} dx$ (without including limits). Parts (b)(i) and (b)(ii) were usually answered correctly although a minority of students used the incorrect formulae for tension and elastic potential energy. In part (b)(iii), most students

included terms in kinetic energy, gravitational potential energy and elastic potential energy but many did not include elastic potential energy at both A and the point when its speed was 0.8 ms^{-1} . Another common error was in using the same variable for the height that the particle moved (in the potential energy gained) and for the extension when the particle's speed was 0.8 ms^{-1} .

Q7.

This question was answered successfully by virtually all students.

Q8.

This question was designed to discriminate between grades A and A* and it was common for students who scored very well in the rest of the paper to find this question challenging. Many students were successful in part (a) but some students considered the initial Elastic Potential Energy in the string rather than the initial tension. Part (b) was also completed well by many students although a number forgot the square in their elastic potential energy term. To solve part (c) of this question, students needed to consider four terms, which were the initial elastic potential energy, the work done by friction, the change in potential energy of particle B and the final elastic potential energy. Many students forgot that the string had a non-zero elastic potential energy when particle B was at rest. Relatively few students considered the work done by the frictional force which was the friction acting on particle A (31.36N) multiplied by the distance A had moved (0.46m). A few students incorrectly used $v^2 = u^2 + 2as$, ignoring the fact that constant acceleration formulae cannot be used with elastic strings.

Q9.

Candidates answered this question well. Virtually all candidates used energy as required and scored all 8 marks, although, in part (c)(i), some subtracted the change in potential energy from the initial kinetic energy. A few ignored part (c)(i) and only answered part (c)(ii).

Q10.

In part (a), most candidates assumed that conservation of energy was required. Unfortunately, some tried to use $v^2 = u^2 + 2as$. In part (b) candidates had difficulty in identifying their 'u' and their 'v', often not stating whether they were considering the tension at point A or at point B. Sometimes the force equation at A (or at B) was simply given as $T = mg$, with candidates ignoring any circular motion. The fact that T_A and T_B were stated to be in the ratio of 2 : 5 caused major problems. Some stated that $T_A = 2$ and $T_B = 5$ - this was not accepted as it could not produce the correct solution - other candidates used $T_A = 2T$ and $T_B = 5T$, which was valid and often led to a successful solution. A number of candidates used $T = 2T$ and $T = 5T$; if they managed to keep track of which of these Ts were T_A , T_B or the T used in their ratio, this was condoned. A

significant number of candidates incorrectly used $T_A = \frac{5}{2}$, which usually resulted in a negative value for u^2 . In part (c), those candidates who were successful in part (b) found $v^2 = 5ag$. However many gave the ratio $u : v$ to be 9 : 5 rather than $3 : \sqrt{5}$.

Q11.

Parts (a) and (b) were usually answered correctly but in part (c)(i), a number of candidates subtracted the two energies rather than adding them. Part (c)(ii) was usually completed

correctly.

Q12.

In part (a) most candidates realised that they needed to consider the kinetic energy at the lowest point and the change in potential energy moving from P to the lowest point. Methods used to find the change in height between points were interesting, but often not correct. The answer for the change in height was often truncated and this caused inaccuracy in the answer for the maximum speed. In part (b), many candidates correctly found an equation involving T .

Q13.

In part (a) most candidates found the initial EPE correctly, but a number forgot to add on the initial KE to find the energy of the block at A . In part (b)(i), most candidates found the frictional force to be $0.4 \mu g$ but often failed to multiply this by 5.5 to obtain the work done by the frictional force. Other candidates used $v^2 = u^2 + 2as$, ignoring the fact that constant acceleration formulae cannot be used with elastic strings. In part (b)(ii), candidates often used the speed after rebounding to be half the speed of the block before the impact, but when finding the kinetic energy forgot to square the half, when calculating v^2 . The required equation on part (ii) included three terms, KE when leaving B , work done by friction and the EPE in the string when the block came to rest at B . Unfortunately many candidates forgot that the string had a non-zero EPE when the block was at B .

Q14.

This question was also answered well by most candidates. However, a small number of candidates found the potential energy to be $4 \times g \times 5 \cos 20$ rather than $4 \times g \times 5 \cos 70$.

Q15.

Most candidates completed parts (a) and (b) of this question correctly. In part (c), a number of candidates tried to use power = force $\times$ velocity and multiplied by 2, the speed of the water, instead of dividing by 60, the time in seconds.

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Q16.

In part (a), most candidates used two kinetic energy terms and either one or two potential energy terms. They quickly found that $8v^2 = 2ag$, but some could not solve this to find a value for v .

In part (b), most candidates realised that the maximum and minimum values of the tension were at the lowest and highest positions of the bead, but there were a number of sign errors in the resolving necessary to find the tensions. A significant number of candidates found the maximum

tension to be $\frac{m(5v)^2}{a} = mg$ but did not substitute the value of v which they had found in part (a).

Q17.

Many candidates were unable to show convincingly that the work done in stretching the string

was $\frac{\lambda e^2}{2l}$. An integration using a variable, usually x , was required.

Part (b)(i) was usually answered correctly. In part (b)(ii), most candidates appreciated that they needed to consider gravitational potential energy and elastic potential energy. However, many candidates did not appreciate that the string was still stretched when the 4 kg block was next at rest. Often candidates used terms in kinetic energy, gravitational potential energy and elastic potential energy and then equated the speed to zero. In part (b)(iii), the extension x when the speed is at a maximum was usually found by using $T = mg$. A few candidates used techniques involving the maximising of v^2 ; impressively, sometimes these were totally successful.

Q18.

Part (a) was usually answered correctly, but in part (b) a number of candidates did not use 7 metres to be the difference in Kim's height. A few candidates did not add the 116 J (from part(a)) to find the new kinetic energy.

Q19.

In part (a), most candidates considered the kinetic energy at the lowest point and the potential energy at the highest point. Unfortunately many equated these energies and thus obtained $u = 2\sqrt{ag}$. Noticing the result required, these then just changed their results to $u > 2\sqrt{ag}$; this was not accepted.

In part (b)(i), many candidates attempted to find at least one of the two equations required.

Common errors seen were in the conservation of energy equation, where the 'height' in $\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mga(1 + \sin \theta)$ was often incorrect, and in the resolving equation $\pm R = -mg \sin \theta + \frac{mv^2}{r}$, where the $\sin \theta$ term was often incorrect. Few candidates obtained the correct value for R .

In part (b)(ii), virtually all candidates equated R to zero but rarely solved their resulting equation.

Q20.

Part (a) was answered well by most candidates. Most candidates answered part (b) correctly, although a few tried to use tension rather than elastic potential energy. In part (c), most candidates appreciated that they needed to consider kinetic energy, work done and elastic potential energy, although a few also tried to include gravitational potential energy. Many candidates did appreciate that the work done was force $\times$ distance but did not realise that this distance was not the extension of the string.

Q21.

Most candidates appreciated that they needed to consider kinetic energy, gravitational potential energy and elastic potential energy. However the extension in the cord was not equal to the height dropped by the bungee jumper and this caused difficulty to many candidates.

In part (b) most candidates gave a reasonable explanation why x had to be greater than 22.

In part (c) most candidates knew that $v = 0$ when x had a maximum value, but only the better candidates appreciated that, since x had to be greater than 22, x had to be the larger of their two solutions. This statement was required for full credit to be given.

In part (d) (i) the extension x when the speed is at a maximum was found either by maximising $5v^2$ or by using $T = mg$, and candidates used these two approaches in equal numbers with equal success. Part (d) (ii) was answered well.

Q22.

Candidates also answered this question well. Virtually all candidates considered energy as required, although, in part (c)(i), some subtracted the initial kinetic energy from the change in potential energy. Others ignored part (c)(i), and only answered part (c)(ii). Most candidates gave an appropriate answer in part (d), stating either that the rock is a particle or that there is no air resistance.

Q23.

This question was also answered well. However, in part (b)(i), a number of candidates found the maximum height which the stone reached and used this to find the kinetic energy when it hit the ground, rather than simply using the fact that the kinetic energy on impact is the initial kinetic energy plus the change in potential energy. Part (b)(ii) was also well answered and most, in part (b)(iii), gave the correct modelling assumptions that (1) the stone is a particle, and (2) there is no air resistance.

Q24.

Most candidates answered part (a) correctly. In part (b), a significant number did not read the question carefully to see that an *energy* method was required. Thus the *only* acceptable method was to use $F_s = \text{change in kinetic energy}$.

Q25.

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Many candidates experienced difficulty with this question. In part (a), candidates rarely used the fact that the acceleration of the bungee jumper was zero when the tension in the cord was equal to the gravitational force. Candidates tried to use energy considerations in part (b)(i), but did not use the correct difference in height for potential energy. There was a significant amount of 'creative' algebra to obtain the printed equation. In part (b)(ii), most candidates solved the given equation quickly, and many then subtracted this from 65 to find the distance between the bungee jumper and the ground. The three significant figure solution to the quadratic resulted in a two significant figure answer which was not penalised. This rarely concerned candidates; only the better ones found the four significant figure solution to the quadratic in order to obtain a three significant figure answer.

Q26.

Many candidates showed that they knew how to answer this question. However, when considering the slide at the water park, many of them misread the phrase "inclined at an angle of 30° to the vertical" and based their calculations on the more commonly-seen slope of 30° to the horizontal.

In part (b), the question asked for both the kinetic energy *and* the speed, and some candidates only gave one of these. In part (c), candidates often gave a modelling assumption which had been quoted in the question: for example, that the plane is smooth. Only assumptions not given in the question were accepted, such as that Anne is a particle.

Q27.

Parts (a) and (b) of this question were answered well. In part (c), many complicated methods were used to find the work done; often speed was considered and it was common for the work done, exactly 48 J from part (a), to become a value fairly close to 48 J. Most candidates found the frictional force to be $5g\mu$, but relatively few candidates used work done by friction to be $5g\mu$ multiplied by the distance moved, which was 2 m.

Q28.

Candidates found this question straightforward and answered it well

Q29.

Parts (a) and (b) were done well. In part (c), many candidates did not read the question carefully. All too often, only the speed was found, ignoring the requirement to give the kinetic energy.

Q30.

Most candidates answered part (a) well. In part (b), most candidates attempted to find the final kinetic energy. Unfortunately, a few candidates found an expression for the final speed without showing the result of the intermediate calculations, and merely wrote the result as given in the question, 3.66 m s^{-1} . Such candidates were penalised. Candidates should appreciate that a question requiring them to “Show that ...” means they cannot simply write down the answer. In part (c), candidates used a variety of methods, generally based on energy, to show that the particle could not reach A.

Q31.

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Part (a) tested work done = $\int F dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line y

= $\frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for

example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and $x = e$; or work done = maximum force $\times$ half the distance moved, without a clear justification.

Part (b)(i) and (ii) were completed well. In part (b)(iii), most candidates attempted to conserve kinetic energy, potential energy and elastic potential energy, but often the signs

of the three terms in the equation $\frac{1}{2}mv^2 - 245 + 10g = 0$ were incorrect.

Q32.

Part (a) tested work done = $\int F dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and $x = e$; or work done = maximum force $\times$ half the distance moved, without a clear justification.

Part (b)(i) was completed well, as was part (b)(ii), although a number of candidates did

write down $\frac{150 \times 0.9^2}{2 \times 1.2} = 11.25$, which evidently it is not. In part (b)(iii), many tried to use the conservation of kinetic energy, potential energy and elastic potential energy. The printed result enabled many of those who obtained an incorrect answer to correct their working. Commonly, this was because they had used $\frac{\lambda(0.9 - x)^2}{2l}$ instead of $\frac{\lambda(0.3 - x)^2}{2l}$ for the elastic potential energy. Part (c) was completed well, but some candidates did not explicitly exclude $x = 0$.

Q33.

Parts (a) and (b) were answered very well. Some omitted part (c) while others stopped after finding the change in energy.

Q34.

In part (a), many candidates used conservation of energy, not to find the kinetic energy when the string became taut, but to find the velocity when the string became taut. Then they used $KE = \frac{1}{2}mv^2$ to find the kinetic energy when the string became taut. The simpler method of using conservation of energy with gain in kinetic energy equalling the loss in potential energy was rarely seen; this of course immediately gave the required kinetic energy. In part (b), most candidates used the change in potential energy to be the change in EPE and readily found λ .

Q35.

This was found to be a good introduction to the paper. In part (b), instead of using conservation of energy, a few found the maximum height that the box reached in its motion. However, because they rounded or truncated this height, these candidates did not usually obtain the exact answer of 1720J. Most candidates gave simple modelling assumptions in part (d) although a few gave one response in two different ways.

Q36.

Parts (a) and (b) caused few problems for the vast majority of candidates. In part (c) it was common to see the frictional force to be $5g \times 0.4 = 19.6$ N, but this was often not multiplied by the distance moved, 0.5 (m), to find the work done.

Q37.

Parts (a) and (b) of this question were done very well, with many candidates gaining full marks on both parts. Part (c) (ii) was also done well, but some candidates did not show both solutions to the quadratic as part of their working. In questions like this candidates should be encouraged to show both solutions and then select the appropriate solution in the context under consideration. Part (c)(i) was very challenging and there were relatively few correct solutions. The three main problems were:

- Not including the elastic potential energy when released,
- Using an incorrect expression for the extension of the string when forming an expression for the elastic potential energy,
- Using an incorrect distance when forming an expression for the gravitational potential energy.

Q38.

Generally the candidates did very well on this question and a good number scored full marks. Some candidates used constant acceleration equations to find the speed of the stone when it hit the ground and then calculated the kinetic energy using the formula $\frac{1}{2}mv^2$. While this approach was acceptable in this context, it is a better strategy for candidates to use an energy method, as in some contexts constant acceleration equations cannot be applied.

Q39.

Part (a) was done very well, with very few incorrect responses. Part (b) of the question was also done fairly well, with the printed answer helping some candidates. The main issue here was that some candidates did not use the correct distances to find the elastic potential energies. Part (c) was much more demanding. Many candidates were able to calculate the magnitude of the friction force correctly, but far fewer were able to incorporate this correctly into an energy equation.

Q40.

Parts (a) and (b) of this question were done very well by the vast majority of candidates. The number of candidates using constant acceleration equations instead of energy was quite small.

Part (c) was found more difficult. There were some candidates who did not really know how to proceed with the question, and others who were confused about how to calculate the work done against the air resistance. One of the most common incorrect responses was to produce the answer 47.04 J.

Part (d) was answered well by some candidates, but some gave fairly poor and very brief answers.

Q41.

Part (a) was done well, with candidates able to obtain the printed result convincingly. In part (b) there were many successful solutions. However, some candidates assumed the force on the particle to be constant and applied Newton's Second Law followed by the equations of constant acceleration.

Q42.

This question also proved popular, with parts (a) and (b)(ii) showing a sound understanding of the principles of energy and work. There were many interesting solutions to part (b)(i), with the majority realising that the velocity of the particle increased at stages throughout the motion. Often candidates produced the correct shape in their sketch, but labelling of the axes was incomplete.

Q43.

There was an excellent response to this question. The formula for elastic potential energy now seems well known. The printed answer in part (b) helped candidates correct their work when they obtained a mismatch. Only a few candidates failed to use the correct extension in order to find k .

Q44.

A very pleasing response was made to this question. Formulae for potential energy and kinetic energy are well known. In part (b) candidates were sometimes guilty of double-counting energy terms by muddling up what applied where. Those candidates who considered the situation at the start and at the end were generally more successful. A few candidates still quoted $v^2 = u^2 + 2as$ for part (b)(i) despite previous Examiners' reports highlighting this error in the past – no marks are awarded for this. Responses to part (b)(ii) were far better than in the past although sometimes $N - mg \cos \theta$ was seen or, more worrying, the acceleration was treated as 0. Candidates who tried a vector approach were always unsuccessful. Many correct solutions were seen for part (b)(iii). It was pleasing to see that solving an equation to find when $N = 0$ was equally as popular as finding N when $\theta = 60^\circ$. Only a few candidates tried to set up equations with the velocity involved.

Q45.

There was a very good response to this question. The formula for elastic potential energy seemed to be well known and units were almost totally consistent and correct. The printed answer in part (b) helped candidates to correct their work when they obtained a mismatch. In part (c) a mark was often lost for failing to add on the initial 3 cm, if an approach using kinetic and potential energy was used.

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Q46.

Work here showed an improved response in comparison with previous papers, even though the question was non-standard in some parts. Candidates generally found no problem with part (a). It was, however, disappointing to see some candidates use ' $v^2 - u^2 = 2as$ ' in part (b)(i) and, as a result, score nothing.

Many candidates drew a correct force diagram and, as a direct result, many correct answers were obtained in part (b)(iii).

Part (c) was the most demanding part of the question and few candidates gave a fully correct response, although many scored 1 mark for either considering their expression from part (b)(iii) or for finding the normal reaction value when $\theta = 90^\circ$.

Success in part (d) generally depended on how well candidates had followed the story in the question. Most used ' $v^2 - u^2 = 2as$ ' correctly but several correct approaches using work done = change in kinetic energy were also seen.

Q47.

This was the most discriminating question on the paper. Part (a) did not particularly trouble any candidate.

In part (b), candidates who had been taught the $\frac{1}{2} \frac{\lambda x^2}{l}$ and the $\frac{\lambda x}{l}$ formulae were often not able to complete the question successfully. The specification is clear in its expectation. Weaker candidates often got muddled by mixing up energy terms from different points in the motion. It was pleasing to see that where a candidate had drawn clear diagrams the answer followed easily. It is worrying that many did not score two marks on the quadratic equation. Incomplete methods, invalid methods and failing to clearly rule out -7 as a solution were some of the reasons. Part (b)(iii) was scarcely seen correct, as many candidates did not realise that $mg - T = ma$ was required.

Q48.

This vertical circle question prompted the best response to any seen so far. Focusing on this topic for coursework does seem to help.

In part (a), some candidates scored zero by failing to consider forces at C. However, most provided excellent solutions.

In parts (b) and (c), marks were sometimes lost when energy terms from different points got mixed up. Candidates who quoted $k = 2.5$ from memory failed to score.

A good range of improvements was suggested in part (d), with the inclusion of air resistance and friction being the most favoured. It was surprising to see that some candidates did not seem to understand the request "to refine".

Q49.

This question was answered well by the majority of candidates, who were probably helped by the printed answers.

Part (c) caused a few more difficulties for some candidates than the earlier parts. The main errors were to assume that the kinetic energy would be 10 J or to include gravitational potential energy in some way.

Q50.

Parts (a) and (b) of this question were generally done very well by the vast majority of candidates. Parts (c) and (d) caused more problems for the candidates. Quite a number of candidates tried to use equations of motion and constant acceleration equations to answer parts (c) and (d). While some were able to do this successfully, many were not. In part (c), quite a few candidates simply assumed that $F = 35g \sin 40^\circ = 220$. *Candidates were sometimes more successful with part (d), but there were some who used 560 J for the kinetic energy at B instead of 630 J .*

Q51.

Part (a) of this question was generally done very well by the candidates. Part (b) was found to be more challenging. A few candidates did not attempt either parts (b)(i) or (b)(ii), with others having a go at one but not the other. In both parts, some candidates tried to use constant acceleration equations instead of energy methods. The candidates tended to be more successful with part (b)(i) than with part (b)(ii). In part (b)(ii), quite a few candidates formed and solved an energy equation, but did not interpret their result.

Q52.

Most candidates answered this question well, although the weakest did fudge the algebra to obtain the printed result. Few candidates drew a force diagram in part (b), and these relied upon intuition to obtain the required result.

Q53.

This question was generally done very well. Most of the candidates had a good working knowledge of the energy ideas required. Part (c) did cause difficulties for a few candidates who used the wrong energies.

Q54.

Part (a) was done well, with the printed answer helping some candidates. However, some candidates did try to use elastic potential energy in their solutions. In part (b), several candidates struggled to form a correct equation. Those who did normally went on to produce the required equation. Common errors were to omit the gravitational potential energy or to use $L - 2$ when finding the gravitational potential energy. Part (c) was often done well, but there were some candidates who seemed unaware of any of the techniques that could be used to solve a quadratic equation.

Q55.

There were many good attempts at this question. Part (a) was generally done well, with some candidates basing their approach on the use of constant acceleration equations. A few candidates used an incorrect distance and made very little useful progress with the question. In part (b), several candidates seemed to have been helped by the printed result. Some candidates did not justify all the steps that they took. While most candidates were able to solve the quadratic equation without difficulty, it did cause problems for some. Some candidates did not add the natural length to the extension to find the maximum length. Many candidates gave good assumptions, but some quoted assumptions that would be difficult to describe as important, or ones that had been covered in the statement of the question.

Q56.

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The candidates found parts (a) and (b) quite straightforward, with only a few candidates making minor errors. There were more errors in their attempts at part (c). In part (c), some candidates used constant acceleration equations to find the speed and then found the kinetic energy. Some candidates found this part more difficult, usually making errors with the distances that they used in their calculations.

Q57.

The candidates found the last part of this question very challenging, but the earlier parts straightforward. In part (a), a few candidates omitted the factor of $1/2$ from the formula for the elastic potential energy. Very few were unable to find the kinetic energy given in part (b)(i). In part (b)(ii), a few candidates made errors by using the distance in their energy calculations, but the majority of candidates produced the correct value. Most of the candidates made very little progress with part of the question. In (b) (iii) the major stumbling block was not to include the elastic potential energy in their equations. Those who did include the elastic potential energy could make some progress with the solution. There were only a few correct answers to this part of the question.

Q58.

Part (a) of this question was done well by many candidates. A few used 14.7 instead of 29.4 when trying to find the speed. The main difficulty that the candidates experienced was with part (b). Here, many candidates found the work done against the resistance force, but did not attempt to go further to find the force. Several candidates gave the answer 28.4 N, which showed some confusion between force and energy.



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